

Module 3: When Data Is Not Simple

Non-Parametric Tests in Easy Steps

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24th September, 2026

What We Will Do in This Module

- ① Understand when standard tests do not fit
- ② Use Mann–Whitney instead of a t-test
- ③ Use Wilcoxon for before-and-after
- ④ Use Kruskal–Wallis instead of ANOVA
- ⑤ Use chi-square for two categorical variables
- ⑥ Use Spearman instead of Pearson correlation

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- 4 Use Kruskal–Wallis instead of ANOVA
- 5 Use chi-square for two categorical variables
- 6 Use Spearman instead of Pearson correlation

These are called **non-parametric tests**. They work even when the data is skewed, ranked, or from a small sample.

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Non-parametric tests do not assume any particular shape. They work on the **order** of the values, not the values themselves.

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If one household earns 1 crore and another earns 10,000, their ranks are still "top" and "bottom". The extreme value does not distort the comparison.

Before We Begin: Load the Data

```
import pandas as pd
from scipy import stats

df = pd.read_csv("village_survey.csv")

print("Data loaded.")
```

Data loaded.

Same dataset as before. The tool `scipy` contains all the non-parametric tests we need.

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Mann–Whitney asks the same question as the t-test: "*are these two groups different?*" But it makes fewer assumptions about the data.

Test 1: The Code

```
rural = df[df['region_type'] == 'Rural']['income'].dropna()
urban = df[df['region_type'] == 'Urban']['income'].dropna()

U, p = stats.mannwhitneyu(rural, urban, alternative='two-sided')

print("U =", round(U, 0))
print("p =", round(p, 4))
```

Same structure as the t-test. We select two groups, run the test, and print the result. The command is `mannwhitneyu`.

Test 1: Reading the Output

$U = 2802.0$

$p = 0.0002$

- **U** is the test statistic - it measures how separated the two groups are
- **p** tells us how likely that separation is by chance

The p-value is close to zero. So we say: **rural** and **urban** income distributions are significantly different.

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Wilcoxon does the same job as the paired t-test but without assuming normality. It is more honest with small or skewed samples.

Test 2: The Code

```
before = df['wellbeing_score'].dropna().iloc[:50]
after = before + 2 + (before * 0.01)

W, p = stats.wilcoxon(after, before)

print("W =", round(W, 0))
print("p =", round(p, 4))
```

Same before-and-after setup as in Module 2. Only the function changes - from `ttest_rel` to `wilcoxon`.

Test 2: Reading the Output

```
W = 0.0  
p = 0.0000
```

- **W** is the test statistic
- A small **W** means the changes are consistent and in one direction
- **p** tells us whether the change is significant

Here, **p** is essentially zero. So the "after" scores are significantly higher than the "before" scores.

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Kruskal–Wallis is the non-parametric version of ANOVA. It tells us whether any of the groups differ, without assuming normality.

Test 3: The Code

```
groups = [  
    df[df['region_type'] == 'Rural']['wellbeing_score'].dropna(),  
    df[df['region_type'] == 'Urban']['wellbeing_score'].dropna()  
]  
  
H, p = stats.kruskal(*groups)  
  
print("H =", round(H, 2))  
print("p =", round(p, 4))
```

The star symbol * unpacks our list of groups into separate arguments. This is a small Python trick - just use it as shown.

Test 3: Reading the Output

```
H = 21.81  
p = 0.0000
```

- **H** is the test statistic, similar to F in ANOVA
- **p** tells us whether the groups differ significantly

The p-value is essentially zero. So the two region types have significantly different wellbeing distributions.

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Chi-square asks: "Is the pattern of counts different from what we would expect if the two variables were independent?"

Test 4: The Code

```
table = pd.crosstab(df['region_type'], df['land_ownership'])  
print(table)  
  
chi2, p, dof, expected = stats.chi2_contingency(table)  
  
print("chi-square =", round(chi2, 2))  
print("p =", round(p, 4))
```

crosstab builds the table of counts. chi2_contingency runs the test.

Test 4: Reading the Output

land_ownership	No	Yes
region_type		
Rural	66	63
Urban	45	26

chi-square = 0.23
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- **chi-square** measures how far the observed counts are from expected
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Here $p = 0.82$, which is much greater than 0.05. So land ownership and region are **not** significantly related.

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Spearman is Pearson's robust cousin. It measures how well two variables move together in **order**, even if the relationship is curved.

Test 5: The Code

```
rho, p = stats.spearmanr(df['education_years'].dropna(),  
                        df['wellbeing_score'].dropna())  
  
print("Spearman rho =", round(rho, 2))  
print("p =", round(p, 4))
```

Same structure as Pearson. Only the function name changes - from `pearsonr` to `spearmanr`.

Test 5: Reading the Output

```
Spearman rho = 0.44  
p = 0.0000
```

- **rho** is between -1 and $+1$, like Pearson's r
- 0.44 is a moderate positive association
- **p** is essentially zero - this is not by chance

Households with more education tend to have higher wellbeing, and this pattern is significant.

A Note on Sample Size

- Non-parametric tests work especially well with small samples
- But very small samples (fewer than 10) give unstable p-values
- If your sample is very small, report confidence intervals too
- And describe the raw data - counts, distributions, examples
- Statistics cannot rescue a poorly designed study

Non-parametric tests are honest about uncertainty. But no statistical test can make a small sample speak for a large population.

When to Use Which Test

Situation	Standard test	Non-parametric
Two independent groups	t-test	Mann–Whitney U
Same group before/after	Paired t-test	Wilcoxon
Three or more groups	ANOVA	Kruskal–Wallis
Two categories related?	—	Chi-square
Two variables related?	Pearson r	Spearman ρ

Use the right-hand column when data is skewed, ranked, or from a small sample.

Common Mistakes to Avoid

- Using a t-test on very skewed data
- Reporting the p-value but not the median or rank statistics
- Assuming a large p-value means "no effect"
- Running non-parametric tests when the data is actually normal
- Ignoring effect size
- Using small samples to make broad claims

Non-parametric tests are not "lesser" - they are a different tool for a different job. Use them when they fit.

Key Takeaways

- ① Use non-parametric tests when data is skewed, ordinal, or small
- ② Mann–Whitney replaces the t-test
- ③ Wilcoxon is for before-and-after comparisons
- ④ Kruskal–Wallis replaces ANOVA
- ⑤ Chi-square tests whether two categories are related
- ⑥ Spearman is rank-based correlation
- ⑦ Always report the test statistic, the p-value, and the effect

Next: making the results visible.